

COMPOSITE MATERIAL DATASHEETS CORE N636 OVER-EXP. HONEYCOMB

MATERIAL IDENTIFICATION

EU 1E001 - This data is controlled under EU Dual Use Regulation 2021/821, as amended. Authorization may be required for the export of the data outside of the EU.

Short informal name	N636-OX
Commercial name	Eurocomposites ECK-R
Specifications	AIMS11-01-007 (Grade B), Z-17.697, ABS5670A1
Description	Honeycomb Cores, N636 Polyamide Paper (meta and para-aramid), Phenolic Resin Impregnated, Over-expanded.
Version	1.0

REFERENCES

- X00RP1410460 Is. 1.0 "Design Values for N636 Over-expanded Honeycomb to AIMS 11-01-007" Airbus Technical Report.

GRADE B:

PHYSICAL PROPERTIES

Cell size	4.8 mm
Density	32 kg/m ³
Max. Operational Limit (MOL)	135 °C

CTE (1/°C)	Range	-55..20°C	20..70°C	70..135°C
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ELASTIC MODULI

Property	Mean Values (MPa)				B-basis K_B	<i>E-KDF</i>		B-basis Values (MPa)			
	-55°C/Dry	RT/Dry	70°C/W	120°C/W		70°C/W	120°C/W	-55°C/Dry	RT/Dry	70°C/W	120°C/W
E_{33c}	34.0 (*)	34.0 (*)	25.8 (*)	25.0 (*)	0.74	0.757 (*)	0.734 (*)	25.2 (*)	25.2 (*)	19.1 (*)	18.5 (*)
G_{13}	33.2	33.2	26.8	26.2	0.74	0.807	0.786	24.4	24.4	19.7	19.2
G_{23}	66.6	66.6	60.3	56.8	0.74	0.903	0.852	48.9	48.9	44.2	41.7

(*) These values are based upon engineering judgement.

STRENGTH

Property	Mean Values (MPa)				B-basis K_B	<i>E-KDF</i>		B-basis Values (MPa)			
	-55°C/Dry	RT/Dry	70°C/W	120°C/W		70°C/W	120°C/W	-55°C/Dry	RT/Dry	70°C/W	120°C/W
F_{33t}	3.382	3.382	2.562 (*)	2.501 (*)	0.74	0.757 (*)	0.734 (*)	2.482	2.482	1.880 (*)	1.836 (*)
F_{33c}	1.092	1.092	0.865	0.849	0.74	0.792	0.777	0.780	0.780	0.635	0.623
F_{13}	0.530	0.530	0.495	0.470	0.68	0.894	0.842	0.380	0.380	0.340	0.320
F_{23}	0.718	0.718	0.664	0.648	0.70	0.924	0.882	0.520	0.520	0.487	0.459

(*) These values are based upon engineering judgement.

OTHER PROPERTIES

None.